BACHELOR THESIS

Development of a Multi-Agent Pipeline Utilizing Large Language Models for Complex Task Automation

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Motivation & Goals

The Opportunity

LLMs enable **complex task automation** through multi-step reasoning.

Current Challenges

Existing Frameworks

- × Heavyweight and opaque
- Cloud- and large-model–oriented

Local LLMs

- Struggle with agent loops
- Unstable tool usage

Thesis Goals

- ✓ **Lightweight, controllable** multi-agent pipeline
- ✓ **Explicit data flow** and deterministic execution
- ✓ Efficient for **local LLMs**, compatible with CrewAI

Core Architecture Overview

Graph-Based Pipeline

Architecture defined as a directed graph where data flows between discrete processing units.

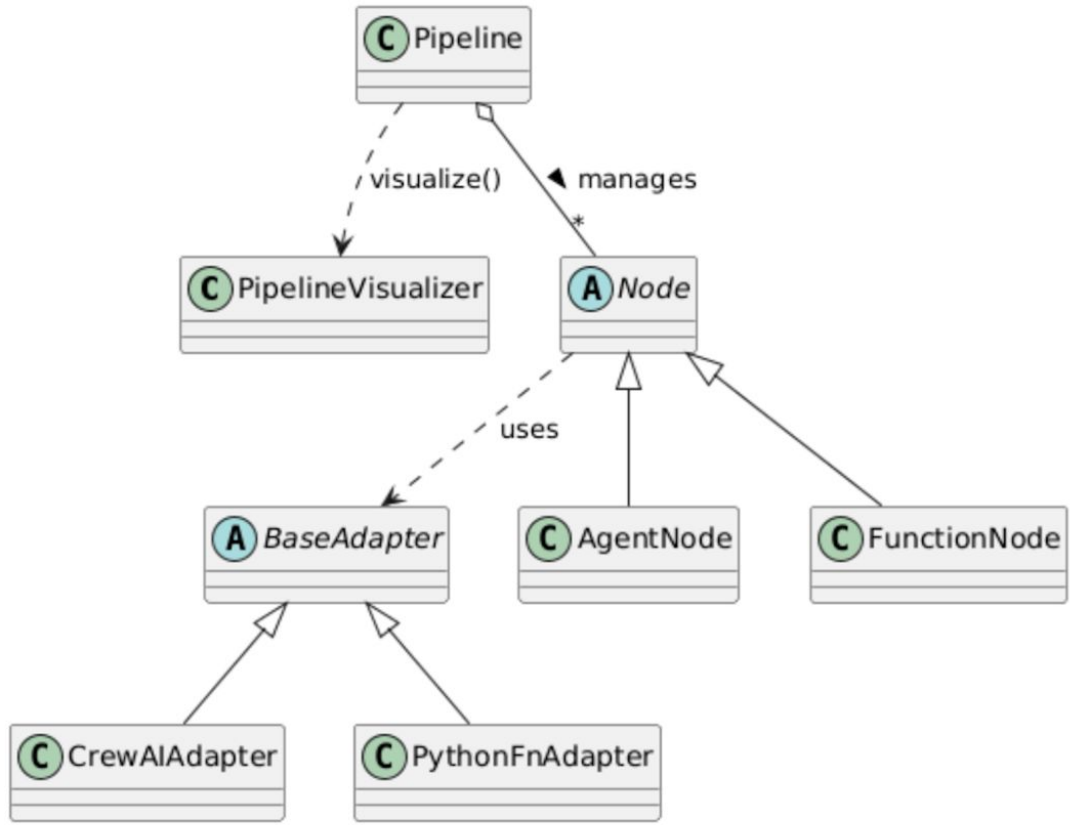
Specialized Nodes

FunctionNode: Deterministic logic

AgentNode: LLM-based reasoning

Adapter Pattern

Decouples orchestration logic from the execution backend, enabling easy swapping of local vs. cloud LLMs.



Orchestration Approach

Control Flow

CORE REQUIREMENTS

Conditional Branching
Dynamic path selection based on intermediate results.

Loops
Refining outputs through controlled repetition.

Nested Pipelines
Modular sub-graphs for complex workflows.

Existing Methods

STATE OF THE ART

CrewAI `Agentic`
Role-based & highly autonomous.
Hard to control with small LLMs

LangGraph `Graph`
Powerful graph structure.
Heavy LLM-centric logic

Proposed Hybrid

THESIS CONTRIBUTION

Graph-Based + Deterministic
Combines rigid structural control with flexible LLM reasoning nodes.

Separation of Concerns
Orchestrator handles flow.
LLM handles content.

"Enables reliable execution on constrained local hardware."

Use Case: Trip Planning Pipeline

Pipeline Inputs

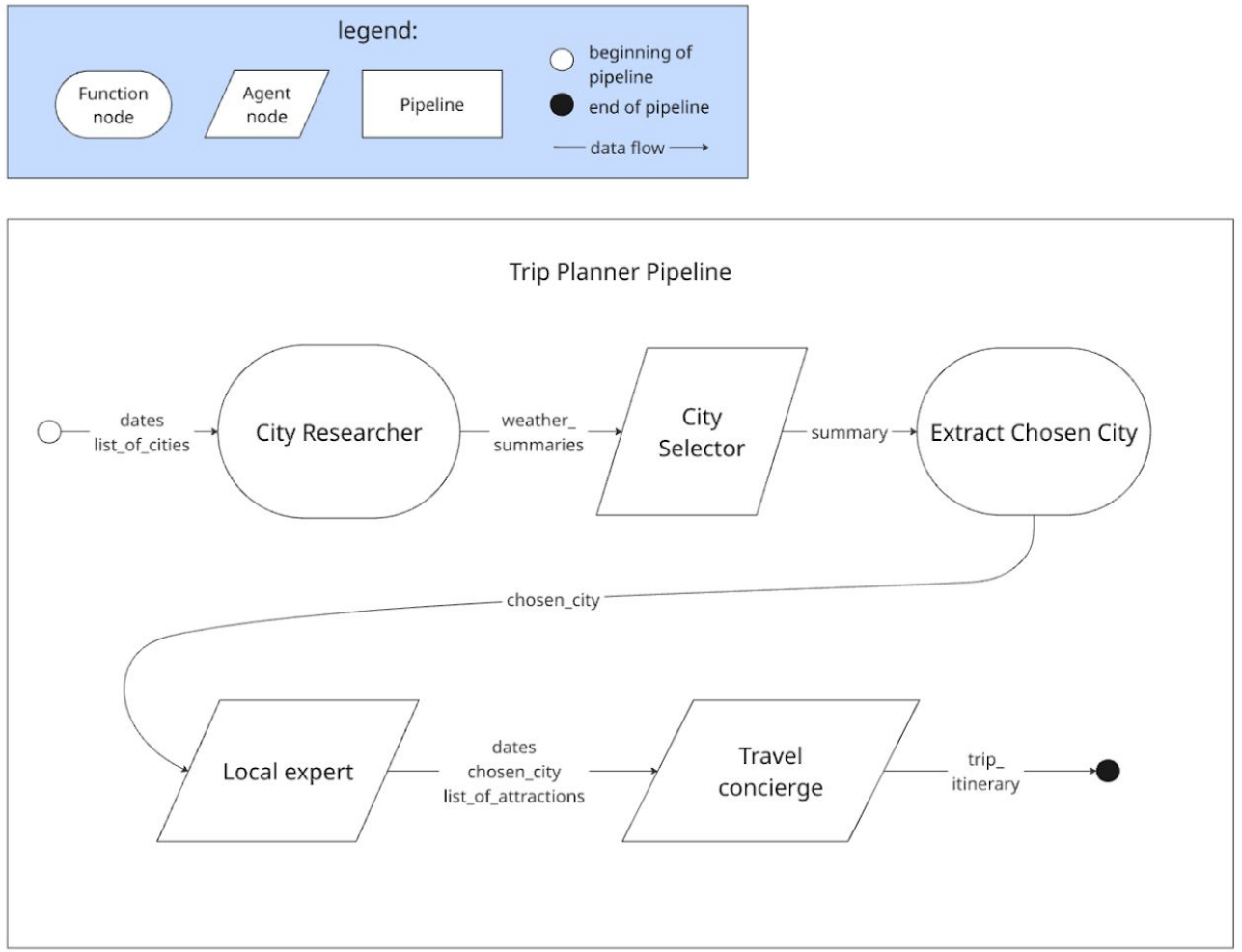
Accepts a structured list of potential cities and a defined travel date range as the initial context for the workflow.

Intelligent Processing

- **Research Loop:** Iterative data gathering
- **Extraction:** Deterministic city validation

Itinerary Generation

Local expert agents curate recommendations before the Concierge agent generates the final day-by-day travel plan.



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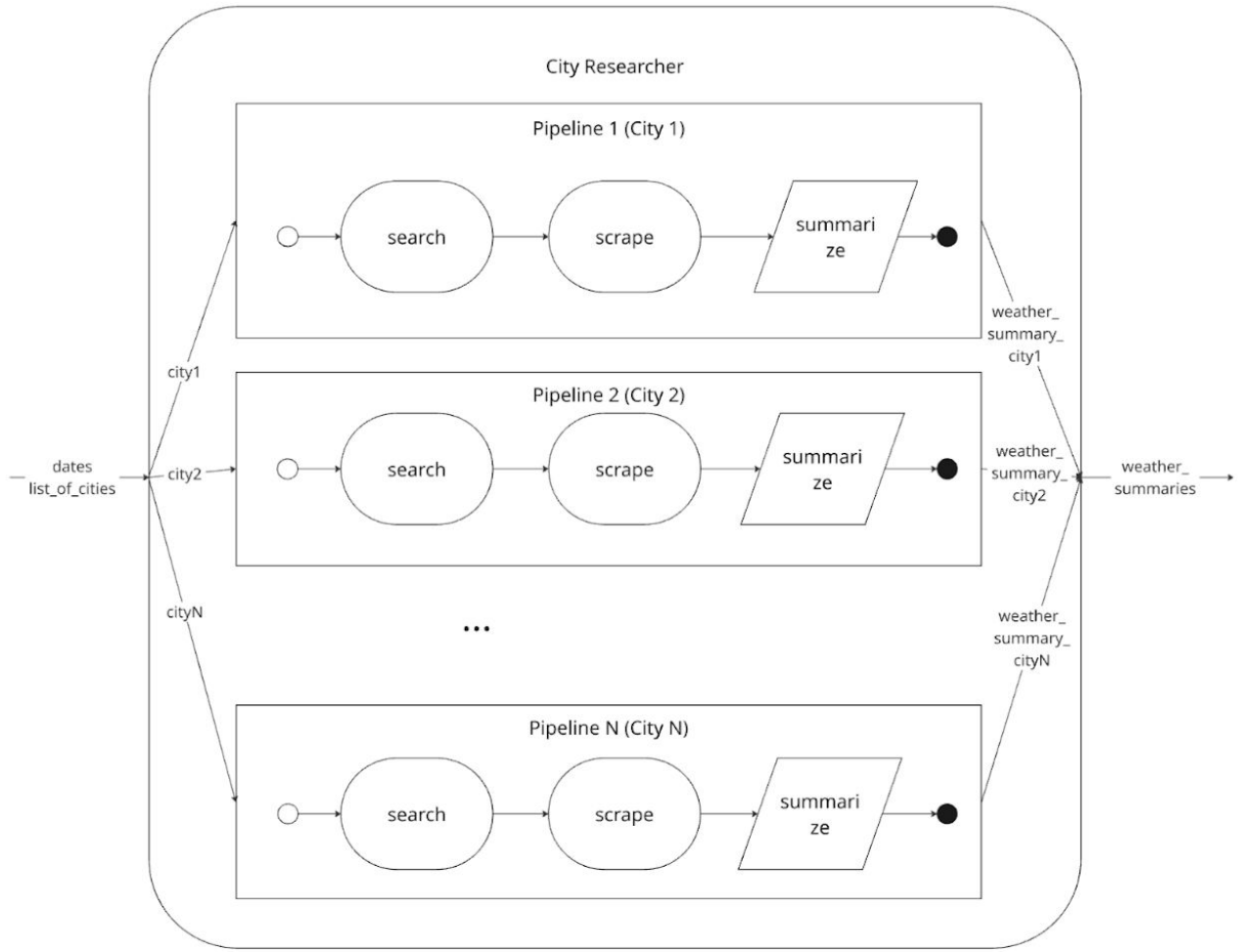

Live Demo

TripPlannerPipeline Execution

Ollama | Llama 3.2

user@local: ~/thesis/demo

● LIVE EXECUTION

```
→ poetry run python3 trip_planner/pipeline.py
[INFO] Initializing Pipeline Graph ...
  ResearchCities → CitySelection → ExtractCity → LocalExpert → TravelConcierge
14:37:30 | INFO | → Running node: ResearchCitiesNode
  Sub-Pipeline:
    DuckDuckGoSearch → Scrape → WeatherSummary
14:37:45 | INFO | → Running node: CitySelectionNode
14:37:52 | INFO | → Running node: ExtractChosenCityNode
14:38:01 | INFO | → Running node: LocalExpertNode
14:38:18 | INFO | → Running node: TravelConciergeNode
[SUCCESS] Pipeline finished in 48.2s
```

Console-Based Interaction

Simple CLI interface accepting natural language or structured arguments. Shows real-time feedback loops.

Transparent Execution

Traceable Logs: Clearly distinguishes between Agent reasoning (LLM) and deterministic Function steps.

Local Performance

Optimized for lightweight models like `Llama 3.2` running on consumer hardware without cloud dependency.

Evaluation Results

Target: Trip Planner (PNA vs CrewAI)

Models: Qwen3:8b, Llama3.2:3b (via Ollama)

Qwen3 + All Tools

SUCCESS

Execution Time — PNA FASTER

249s vs 434s in average

- More stable decision-making due to deterministic pipeline
- CrewAI more verbose but higher overhead

Llama3.2 + All Tools

FAILURE

Both frameworks fail

Agents enter repetitive tool-usage loops

- Infinite loops in tool calling behavior

Unrestricted tools unsuitable for lightweight models

Llama3.2 No Tools

SUCCESS

Execution Time — ~30% FASTER

44s vs 59s

- Deterministic Function Nodes provide required context
- Reduced orchestration overhead improves performance

Llama3.2 Limited

PARTIAL

Tools for Local Expert only

Partial instability in tool-enabled agent

- Final output coherent due to tool-free final agent

Even limited tool usage risky for small models

KEY TAKEAWAY

Explicit orchestration and restricted tool usage are critical for reliable execution on lightweight local LLMs

Limitations

Higher Upfront Effort

Requires explicit definition of nodes and edges. Unlike autonomous agents, the pipeline structure must be **designed manually** before execution.

Reduced Flexibility

Less adaptable than fully autonomous systems. The strict graph structure limits the agent's ability to **deviate or improvise** outside defined paths.

Current Technical Constraints

Small Model Instability

Tool usage and strict format adherence remain unstable on very small quantized models (e.g., < 3B parameters).

Unimodal Processing

Currently limited to text-only processing strings. No support for image inputs or multimodal analysis yet.

Conclusion & Future Work

Key Takeaways

Improved Reliability

Graph-based orchestration significantly reduces hallucination loops common in purely autonomous agents.

Balanced Architecture

Successfully combines the flexibility of LLMs with the predictability of deterministic control flow.

Hardware Efficiency

Optimized for resource-constrained environments, proving viability for local Llama-3.2 execution.

Future Work

RAG Integration

Embed Retrieval-Augmented Generation directly into AgentNodes for knowledge-intensive tasks.

Multimodal Support

Extend pipeline to process image and audio inputs alongside text prompts.

Framework Adapters

Develop adapters to import existing LangChain or AutoGen agents into the graph.

Visual UI

Create a drag-and-drop web interface for pipeline design and real-time execution tracing.